from another host or other external sources it is termed cross infection. Cross infections occurring in hospitals are called nosocomial infections. Iatrogenic infection refers to physician induced infections resulting from investigative, therapeutic or other procedures.

Depending on whether the source of infection is from the host's own body or from external sources, infections are classified as **endogenous** or **exogenous**, respectively.

Endogenous infection

Endogenous infections are acquired from the host himself from the normal flora of the body.

Microorganisms are present in certain areas of the body in all human beings. They are called normal flora. The common areas are Nose, Mouth, Teeth, Throat, Intestine, Urethra, Vagina and Skin (Figure 12.1).

1. When the skin is breached normal flora enters the tissues.
2. When the urethral organisms ascend, they cause urinary tract infection
3. When a patient is treated with antibiotics, normal flora is eliminated and replaced by potential pathogens
4. When the intestine is perforated, normal flora enter the previously sterile body parts
5. Similarly when the pH of the vagina increases potential pathogens occupy the space.

However normal flora helps host against pathogen and benefits the host in many ways

- Normal flora of skin produces fatty acids which inhibit other species
- Intestinal bacteria secrete antibacterial substances (bacteriocins, colicins) and many metabolic products that prevent other species to survive.

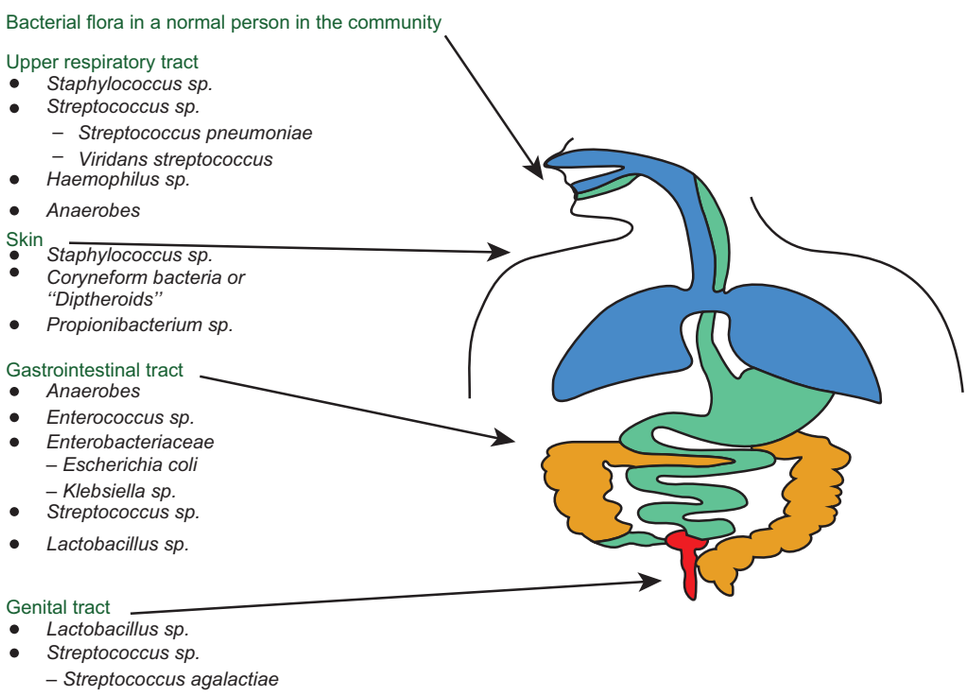


Figure 12.1: Microorganisms present as normal flora
(There are many organisms in a site. Only few are listed)

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